Research Article

Coach WAR: A Counterfactual Approach to Evaluating NFL Head Coach Impact in the Modern Era

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Abstract: Evaluating head coach performance in the National Football League presents unique challenges because of the complex interactions among coaching decisions, player talent, and organizational factors. This paper introduces a novel counterfactual framework for measuring head coach impact by predicting how each team would have performed under a replacement-level coach with league-median characteristics, then comparing this baseline to actual results. This difference represents coaching Wins Above Replacement (WAR), the wins directly attributable to that coach's performance. Analyzing nearly 1700 team seasons from 1970 to 2024 with 368 features spanning draft history, injuries, rosters, team performance, and coaching histories, we employ XGBoost regression to isolate coaching excellence from roster quality and organizational context. The model achieves R^2 of 0.821, with coaching-specific features accounting for 49% of total model importance. Results reveal substantial coaching impact, with elite coaches averaging +2.4 WAR per season and poor coaches averaging -3.2 WAR. Analysis of offensive versus defensive coaching backgrounds reveals a dramatic historical shift: defensive coaches dominated the 1970s (+0.92 win advantage, p=0.0001), but offensive coaches now hold a +0.41 win edge in the 2020s. Year-to-year persistence analysis demonstrates scheme obsolescence, with offensive coaches showing 20% reduced persistence in multi-year performance compared to defensive coaches. This framework provides NFL organizations with objective, context-adjusted coaching evaluation comparable to analytical rigor applied to player personnel decisions.

Keywords: coaching analytics, wins above replacement, counterfactual analysis, NFL, sports analytics

1 Introduction

Evaluating head coach performance in the National Football League presents unique challenges because of the complex interactions among coaching decisions, player talent, and organizational factors. Traditional metrics often conflate team success with coaching quality, failing to isolate a coach's actual contribution to the team. Prior research on NFL coaching evaluation has primarily focused on production efficiency models Clement and McCormick (1989); Hadley et al. (2000) and managerial lifecycle effects Goff and Wisley (2006). However, these studies have not attempted to isolate the coaching impact using counterfactual frameworks that control for roster quality and organizational context.

This paper introduces a novel counterfactual framework for measuring head coach impact: it predicts how each team would have performed under a replacement-level coach with league-median characteristics, then compares this baseline to actual results. This difference represents coaching Wins Above Replacement (WAR), the wins directly attributable to that coach's performance. A higher WAR indicates stronger coaching performance, and conversely, a lower WAR indicates poorer coaching performance. By controlling for roster quality, strength of schedule, and organizational resources, this project isolates coaching excellence or deficiency from the talent and context surrounding it.

2 Methods

This project analyzed nearly 1700 team seasons spanning from 1970 to 2024, representing all 32 current NFL franchises. This period was selected as it is typically referred to as "The Modern Era". This project incorporated 368 features spanning multiple data sources into the prediction model. The data collected included draft pick history, injury reports, starter designations, complete rosters, historical team performance, and comprehensive head coach career histories from Pro Football Reference, and salary cap allocation and contract details from Spotrac. Feature counts by category are shown in Appendix A (Table 9). The coach-specific feature set comprised 139 features capturing coaching tenure and previous team performance during previous hiring instances as either an offensive coordinator, a defensive coordinator, or a head coach. Data availability varied by data category. While team performance and roster metrics extended across all seasons, salary cap data was limited to 2011–2024, and player injury data was limited to 2009–2024.

2.1 The Counterfactual Framework

To isolate coaching impact from roster quality, organizational context, and other applicable factors, this project developed a machine-learning-based counterfactual approach to answer the question: what would this team's performance have been if we had replaced the head coach with a league-median alternative while keeping everything else identical?

Specifically, this project depicted replacement-level coaching as a coach with league-median characteristics across all 139 coach-specific features, including career tenure and historical performance metrics. When generating the replacement-level prediction, the project substituted the actual coach's features with these league-median values while holding all other factors constant. This isolated what a replacement coach would be expected to achieve given identical circumstances. Coach WAR is then calculated as the difference between the team's actual winning percentage and this replacement-level prediction.

This approach parallels established WAR methodologies in baseball and basketball Berri et al. (2006); Palmer and Thorn (1984); Smith (2008); Woolner (2002), where player contributions are measured against readily available replacement-level alternatives. Unlike baseball's replacement level (set at a .294 winning percentage, representing the performance of readily available minor league or free agent players Appelman and Cameron (2013); Baseball-Reference.com (n.d.)), this project defines replacement-level coaching as the league median. This choice reflects the coaching labor market's unique structure: NFL head coaches average only 3.2 years of tenure Goff and Wisley (2006), substantially shorter than MLB players' average career of 5.6 years Witnauer et al. (2007). Thus, this Coach WAR measures performance relative to the average NFL head coach rather than a theoretical minimum-quality hire.

2.2 Model Training

This project employed XGBoost Regression Chen and Guestrin (2016) to predict team winning percentage with coach median metrics, selected for its ability to capture complex non-linear relationships and feature interactions while providing robust feature importance metrics. The model was trained on all 365 features with the winning percentage as the target variable. To handle missing values in the dataset, primarily arising from incomplete early-era salary data and roster information, this project used Singular Value Decomposition (SVD) matrix completion Brand (2002), which preserved data structure while imputing missing values based on observed patterns across similar team-years.

To prevent overfitting, this project employed a coach-based cross-validation strategy, ensuring coaches in the test set were entirely unseen during training (81.8% train, 18.2% test). Hyperparameters were selected through a randomized search. Final hyperparameter values are shown in Appendix C (Table 10).

2.3 WAR Calculation

The XGBoost model predicts the average winning percentage (Win_Pct) for each team under the same circumstances with a league median coach. WAR is calculated according to the following equation:

$$\text{WAR} = 16 \times (\text{Win\_Pct}_{\text{actual}} - \text{Win\_Pct}_{\text{predicted}}) \tag{1}$$

For historical comparability, we standardize all WAR calculations to a 16-game season, even for the 17-game seasons (2021+). This ensures coaches from different eras are evaluated on an equivalent basis and prevents artificial inflation of recent WAR values. For 17-game seasons, this represents a 6% conservative adjustment in absolute measurement.

3 Results

The final model achieved an R^2 of 0.821 on the full dataset, demonstrating strong explanatory power for team performance. Feature importance analysis revealed that coaching-specific features accounted for 49% of total model importance, confirming that coaches exert substantial influence on outcomes when controlling for roster quality, financial resources, and competitive context. Appendix D (Table 11) shows the top 20 features by importance.

3.1 Model Validation

To ensure the model generalizes beyond the training data and captures genuine coaching impact rather than memorizing historical patterns, we employed three distinct validation strategies that test different aspects of model robustness. Each strategy addresses a specific concern about model validity and real-world applicability.

The first validation approach was a temporal holdout that withheld the five most recent NFL seasons (2020–2024) as a test set while training on all data from 1970–2019. This strategy tests whether the model can predict outcomes in a fundamentally different era of football, accounting for rule changes, strategic evolution, and shifting league dynamics. We chose this approach to verify the model captures enduring principles of coaching impact rather than era-specific patterns that would fail to transfer to contemporary football. The model achieved test $R^2 = 0.731$ (train $R^2 = 0.824$), demonstrating strong temporal generalization with a train-test gap of 0.093.

The second validation approach randomly withheld 20% of coaches (54 of 270 total) along with all their seasons from the training data. This strategy tests whether the model can evaluate coaches it has never encountered, simulating the practical use case of assessing a newly hired coach with no prior history in the dataset. We selected this approach because real-world application requires evaluating coaches the model has never seen during training. The model achieved test $R^2 = 0.745$ (train $R^2 = 0.816$), with a train-test gap of 0.071. This strong performance on completely unseen coaches demonstrates the model learns generalizable patterns of coaching effectiveness rather than memorizing individual coach trajectories.

The third validation approach specifically tested the model's ability to evaluate coaches who entered the NFL after 2019, representing the most challenging validation scenario: coaches in the modern game with no historical data. We held out 11 coaches who debuted as head coaches in 2020 or later, training on all pre-2020 data plus 80% of post-2020 data from experienced coaches. This approach was chosen to verify the model can evaluate brand-new coaches in the contemporary NFL without requiring extensive track records. The model achieved test $R^2 = 0.771$ (train $R^2 = 0.811$), the highest test performance with the smallest train-test gap of 0.041. This exceptional performance confirms the model successfully isolates coaching impact from contextual factors, making it directly applicable to evaluating current NFL coaches.

The consistent performance across all three validation strategies (test R^2 ranging from 0.731 to 0.771) demonstrates the model captures genuine coaching impact patterns rather than overfitting to training data.

The progressively smaller train-test gaps (0.093 → 0.071 → 0.041) as validation becomes more targeted suggest the model is particularly robust when evaluating individual coaches within their contemporary context. Table 1 summarizes the complete validation results.

Tab. 1: Validation Strategy Overview

Validation Strategy	Training	Test	Train R^2	Test R^2	Gap
Temporal Holdout (2020–2024)	1,523 seasons (1970–2019)	160 seasons (2020–2024)	0.824	0.731	0.093
Holdout Coaches	1,363 seasons (216 coaches)	320 seasons (54 coaches)	0.816	0.745	0.071
New Coaches in Modern Era	1,652 seasons (pre-2020 + 80%)	31 seasons (11 new coaches)	0.811	0.771	0.041

3.2 Football Context and Career Leaders

While single-season WAR ranged from +6.8 to -5.2 (as discussed in section 3.7), career averages provide a more robust assessment of coaching ability across varying contexts. Among coaches with at least 3 seasons coached, the top performers averaged between +1.4 and +2.4 WAR per season. Conversely, the worst coaches with at least 3 seasons ranged from -3.2 to -1.7 WAR per season. Tables 2 and 3 show the best and worst coaches, respectively, by average WAR per season among coaches with at least 3 seasons coached.

Tab. 2: Top 15 Coaches by Average WAR per Season (minimum 3 seasons)

Rank	Coach	Seasons	Total WAR	Avg. Yearly WAR
1	Kevin O'Connell	3	7.1	+2.4
2	Mike Martz	6	13.0	+2.2
3	Mike McDaniel	3	6.1	+2.0
4	Don Shula	27	52.2	+1.9
5	Tom Landry	19	32.5	+1.7
6	John Madden	9	15.3	+1.7
7	Chuck Pagano	6	10.1	+1.7
8	Bruce Arians	8	13.3	+1.7
9	Raymond Berry	5	8.2	+1.6
10	Kevin Stefanski	5	7.6	+1.5
11	Kyle Shanahan	8	12.0	+1.5
12	John Ralston	5	7.0	+1.4
13	Red Miller	4	5.6	+1.4
14	Joe Gibbs	16	22.3	+1.4
15	Jim Caldwell	7	9.6	+1.4

The reigning coach of the year, Kevin O'Connell, currently leads all coaches with an average WAR of +2.4 per season across three seasons with Minnesota (2022–2024). Mike Martz (+2.2 wins per season over 6 seasons) and Mike McDaniel (+2.0 wins per season over 3 seasons) round out the top three, both demonstrating innovative offensive schemes that consistently outperformed roster expectations.

The career leader board validates the framework's ability to identify coaching excellence across eras. Legendary coaches Don Shula and Tom Landry appear prominently, with Shula averaging +1.9 WAR per season across an unprecedented 27 seasons (total career WAR of +52.2) and Landry posting +1.7 WAR

Tab. 3: Bottom 15 Coaches by Average WAR per Season (minimum 3 seasons)

Rank	Coach	Seasons	Total WAR	Avg. Yearly WAR
1	Pat Shurmur	4	-13.0	-3.2
2	Matt Eberflus	3	-8.8	-2.9
3	J.D. Roberts	4	-10.4	-2.6
4	Scott Linehan	3	-7.4	-2.5
5	Steve Spagnuolo	3	-6.3	-2.1
6	June Jones	3	-6.2	-2.1
7	Paul Wiggin	3	-6.2	-2.1
8	Bruce Coslet	8	-15.9	-2.0
9	Abe Gibron	3	-5.9	-2.0
10	Rod Marinelli	3	-5.9	-2.0
11	Joe Bugel	5	-9.4	-1.9
12	Mike Riley	3	-5.5	-1.8
13	Mike McCormack	5	-9.2	-1.8
14	Kay Stephenson	3	-5.2	-1.7
15	Dan Henning	7	-12.0	-1.7

per season across 19 seasons (total +32.5 WAR). These sustained performances across coaching legends demonstrate the validity of this approach in calculating coach WAR.

At the opposite extreme, the poorest-performing coaches demonstrated persistent struggles across multiple seasons. Pat Shurmur recorded the lowest average at -3.2 WAR per season across 4 seasons (total -13.0 WAR). Matt Eberflus's -2.9 WAR per season ranked the second-worst all-time.

The spread between elite and poor career performance is substantial: approximately 5.5 wins per season separates coaches at the 95th percentile (roughly +2.1 wins per season) from those at the 5th percentile (roughly -2.4 wins per season). For NFL franchises, this represents the vital difference between perennial playoff contention and chronic rebuilding.

3.3 Coach WAR in Aggregate

Taking the average WAR per season over a coach's career removes the temporal aspect of WAR accumulation. To illustrate this temporal component, Figure 1 shows the WAR accumulation trajectory for three standout coaches: Kevin O'Connell, Don Shula, and Matt Eberflus.

Although Kevin O'Connell has a higher average WAR per season than Don Shula, at this point in their careers (3 seasons in), Don Shula possessed a distinctly higher average WAR per season ($\sim$3.3 vs. 2.4). Don Shula's 7th, 8th, and 20th–24th seasons heavily impacted his WAR trajectory for his career. It also becomes apparent that Matt Eberflus's first two seasons, while WAR negative, were surpassed by his 3rd season.

Figure 2 illustrates the relationship between coaching longevity and WAR performance by plotting coaches based on average WAR per season and career length (measured in seasons coached).

While the quadrant breakout based on medians is largely illustrative, a few takeaways become clear. The upper-right and lower left quadrants show a clear positive relationship between coaching quality and career length. While this should be no surprise—since it makes sense for successful coaches to be retained—it does validate market efficiency. However, the number of coaches with sustained careers of 5+ seasons despite averaging near-zero or negative WAR (upper-left quadrant) represents a market inefficiency, as organizations historically retained coaches who demonstrably cost wins. Isolating these long-tenured, poor-performing coaches for potential replacement is a primary use case for this Coach WAR analysis.

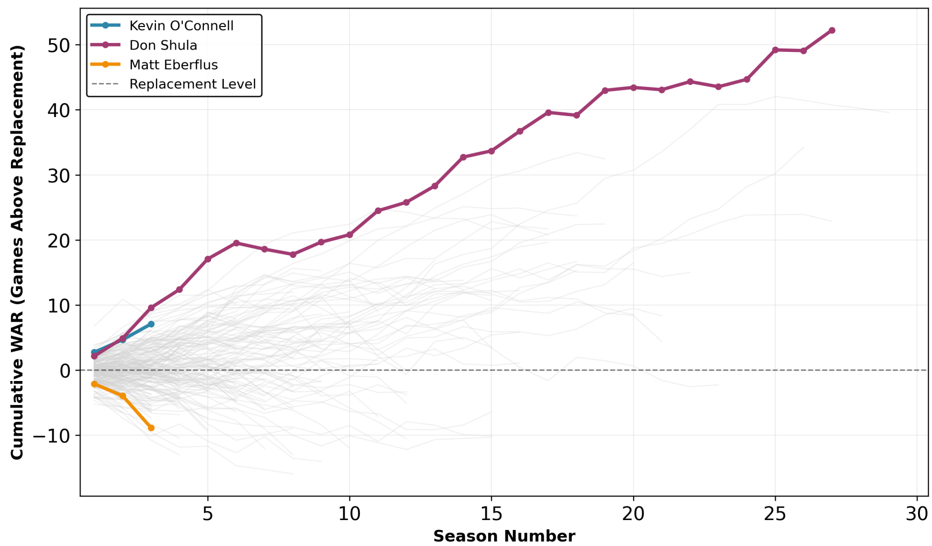

Fig. 1: Kevin O'Connell, Don Shula, and Matt Eberflus Coach WAR Trajectories

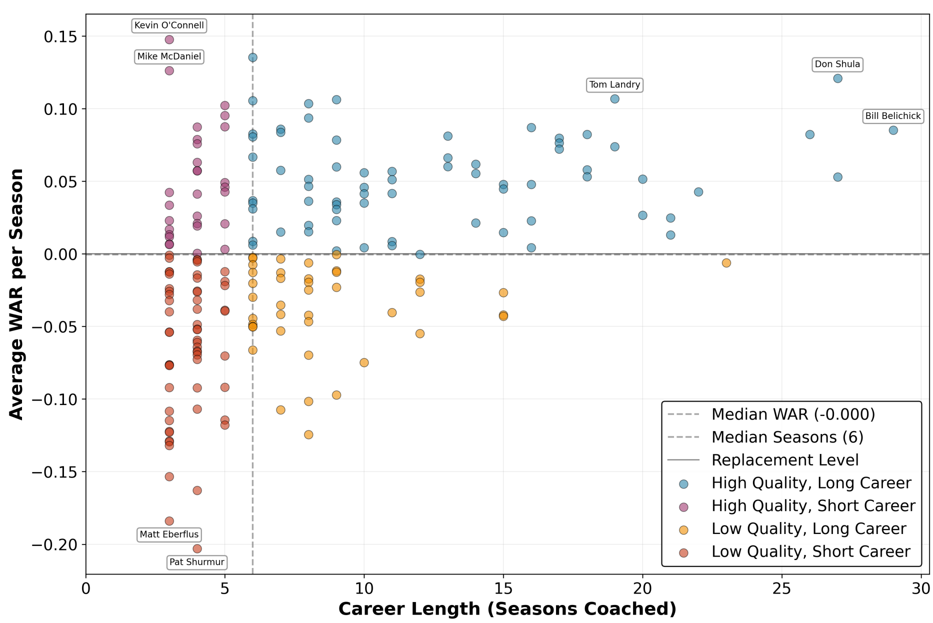

Fig. 2: Coach WAR Career Distributions

3.4 Current Coach Analysis

Figure 3 shows the average career WAR per season for 2024 coaches against their career length. Blue markers indicate coaches who were retained into the 2025 season, and red markers indicate coaches who were fired.

These averages, however, are not the whole story. To dig deeper into the firing decisions of coaches, it's necessary to isolate their performance in the 2024 season. Figure 4 below shows the cumulative WAR trajectories for these coaches, and Figure 5 below highlights each coach's 2024 WAR.

Figure 5 shows how much the firing of Mike McCarthy stands out from that of other fired coaches. With a 2024 WAR of +1.9, Mike McCarthy ranked in the top 25% of NFL head coaches in the season. In

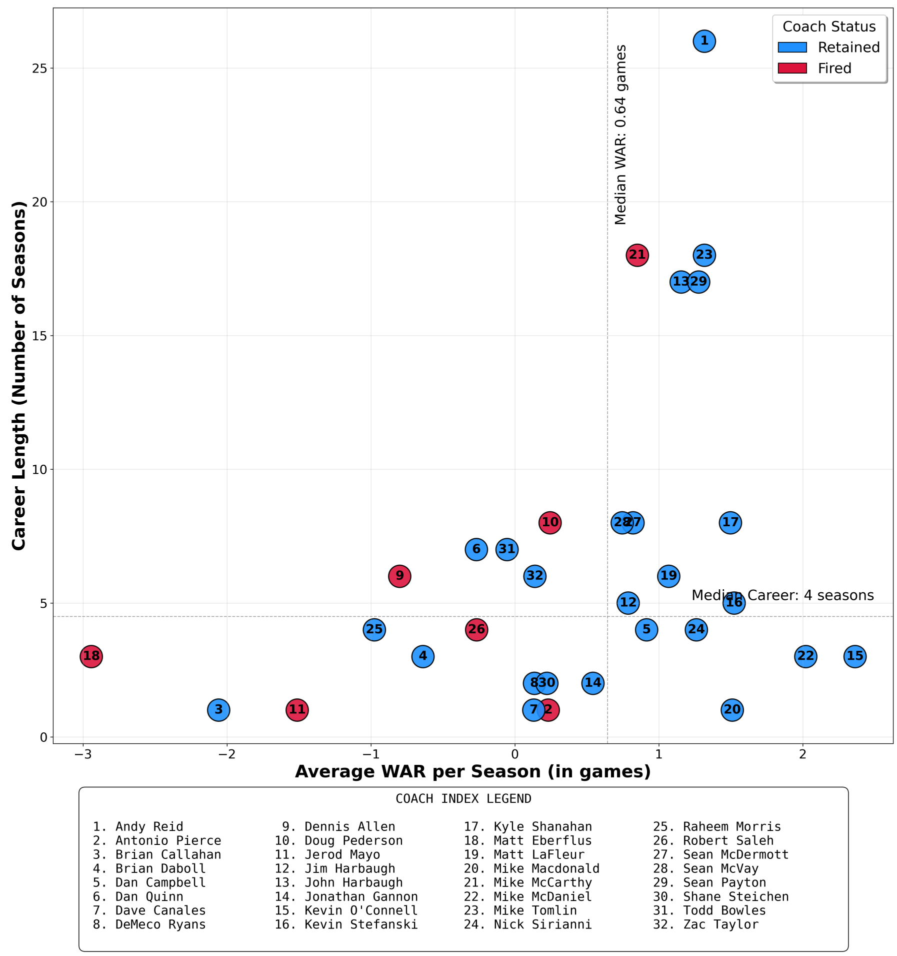

Fig. 3: 2024 Head Coach Average WAR per Season versus Career Length

contrast, four of the seven fired coaches were in the bottom 5 for 2024 Coach WAR. Brian Callahan was the lone exception (a bottom 5 coach who was not fired), but he has since been fired in the 2025 season[1].

3.5 Offensive versus Defensive Coaches

One debate that will likely never be satisfied is whether offensive or defensive coaches are more successful. For this analysis, coaches were classified as offensive or defensive based on their experience as coordinators or position coaches prior to becoming a head coach. Offensive backgrounds included offensive coordinators and offensive position coach roles (quarterbacks, wide receivers, etc.), and defensive backgrounds included defensive coordinators and defensive position coach roles (linebackers, secondary, etc.). This project analyzed Coach WAR by coach type (offensive, defensive, or other/indeterminate) to identify whether coaches of a specific background outperformed other coaches. Figure 6 shows the average coach WAR trajectory for coaches by type for the first 15 seasons of a career as head coach. Table 4 contains Mann-Whitney U test results.

1 Analysis conducted October 2025 on 1970–2024 data; the partially complete 2025 season was excluded from this analysis

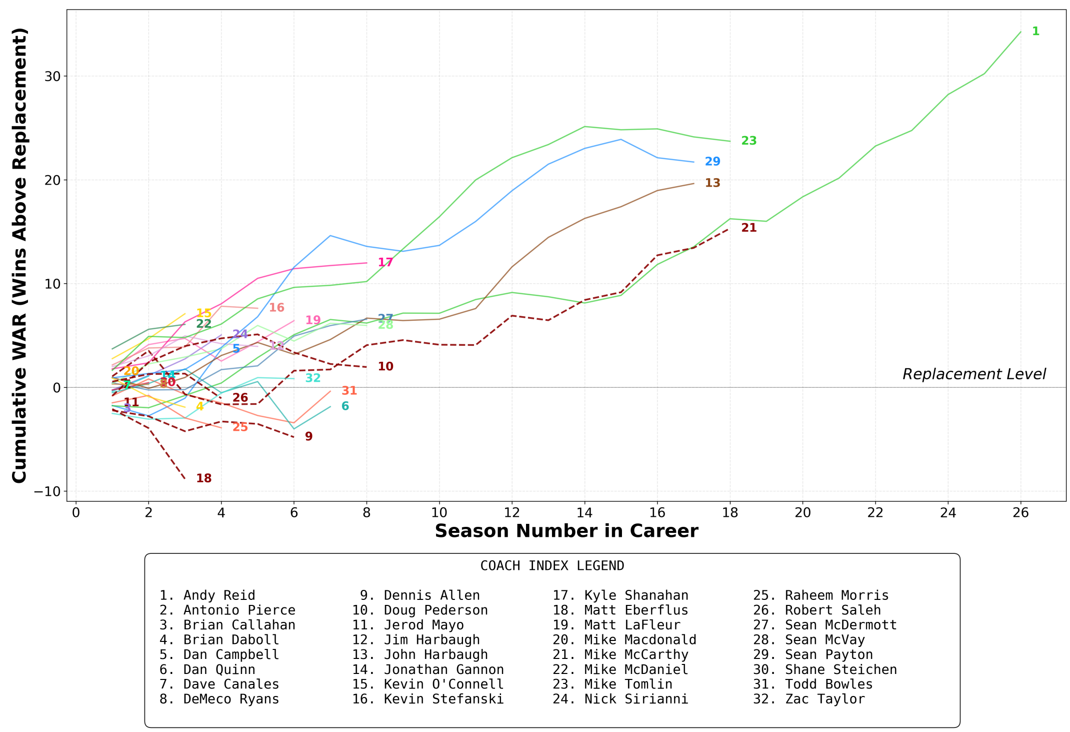

Fig. 4: Coach WAR Trajectory for 2024 Head Coaches

Tab. 4: Mann-Whitney U Test comparing Offensive and Defensive Coaching Backgrounds

Background	Coaches	Seasons	Mean WAR	Median WAR
Defensive	112	639	+0.243	+0.296
Offensive	154	860	+0.088	+0.037
Difference	—	—	0.154	-0.259
Mann-Whitney U Test				
Test Statistic	U = 260,162			
p-value	0.078			
Interpretation	Marginally significant			

This data provides evidence that, at least through the lens of circumstance-adjusted performance, defensive-minded head coaches hold a slight edge on average in the modern era (1970+). However, the gap between defensive and offensive coordinators' performance is less pronounced than the 15-year cumulative totals suggest, as the meaningful divergence doesn't begin until season 4. Across the first three seasons, both groups track nearly identically, indicating no statistically significant difference in how coordinators from either background navigate their initial head-coaching years. The separation emerges in season 4, when defensive coaches accumulate WAR faster than offensive coaches. This occurs again in seasons 10–15, which yield the majority of the difference in WAR. The gap in WAR accumulation in the later stages of careers suggests that offensive coaches may struggle to maintain performance over time, potentially caused by maintaining an outdated scheme / not continuing to innovate with the league.

Coaches from other backgrounds, including special teams coordinators and those whose experience could not easily be classified, underperform both groups initially, averaging slightly negative cumulative WAR throughout the start of their careers. However, by season 5, they begin to accumulate WAR at a similar

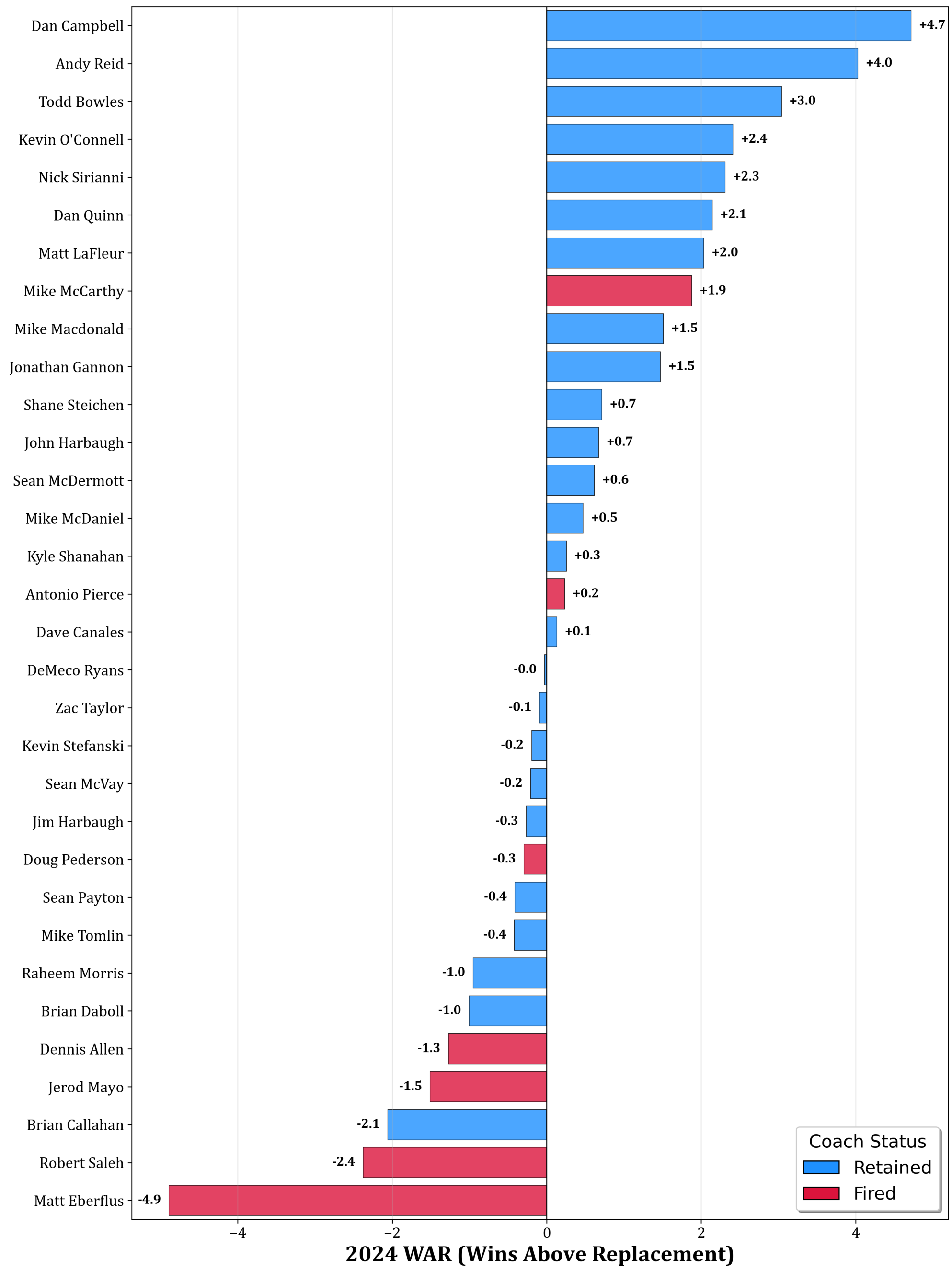

Fig. 5: 2024 Coach WAR for 2024 Head Coaches

rate. This could be explained by the fact that some "other" coaches are less experienced than traditional coordinators, and it takes time to acclimate to the NFL.

But how does the evolution of the game of football play into this analysis? To answer this question, this project segmented the data by decade to isolate the average coach impact for each coach type over time. The results are shown in Table 5 below. Mann-Whitney U-Test results are shown in Table 6 for each decade.

The NFL's evolution from a defense-dominated to offense-dominated league is clearly reflected in coaching success patterns across eras. In the 1970s, defensive-minded head coaches averaged +0.57 WAR per season compared to -0.35 for offensive coaches, a +0.92-win gap favoring defensive backgrounds (p=0.0001). By the 2000s, offensive coaches had pulled even (+0.14 vs. +0.08), and in the 2020s, they've established

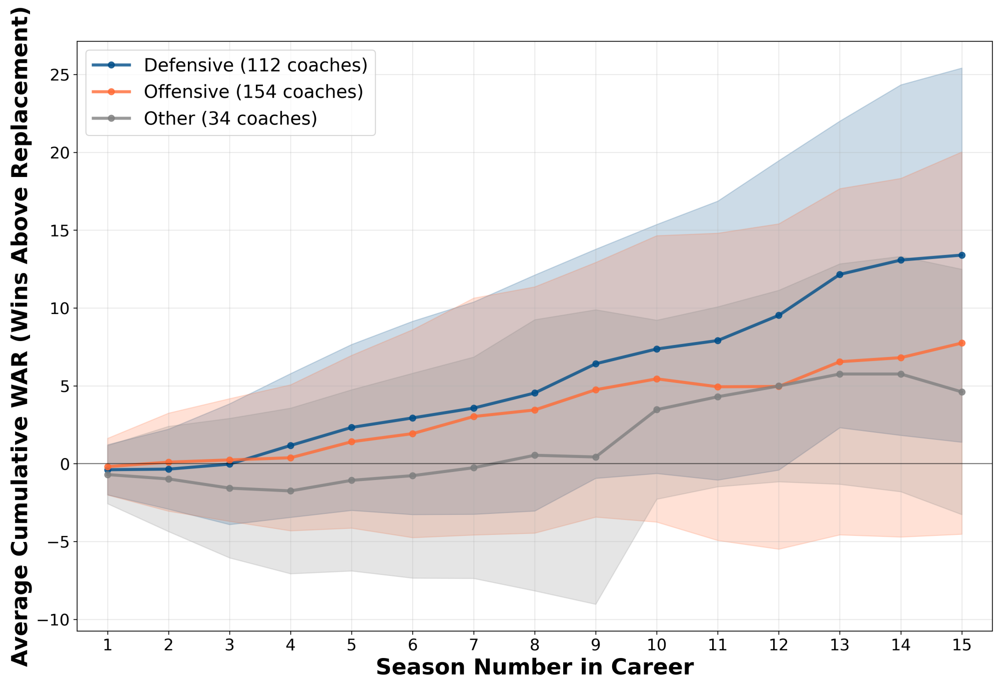

Fig. 6: Average Cumulative Coach WAR by Coach Background (First 15 Seasons)

Tab. 5: Average Coach WAR per Season by Coach Background and Decade

Background	1970's	1980's	1990's	2000's	2010's	2020's
Offensive	-0.35	+0.42	-0.14	+0.14	+0.25	+0.57
Defensive	+0.57	+0.46	+0.19	+0.08	+0.53	+0.16
Other	+0.02	-0.11	+0.51	+0.04	+0.43	-0.07
Delta, 2020's vs 1970's						
Offensive	+0.92					
Defensive	-0.41					

Tab. 6: Mann-Whitney U Test comparing Coaching Backgrounds by Decade

Metric	1970's	1980's	1990's	2000's	2010's	2020's
Sample Size (O/D)	38/24	32/19	38/25	42/33	44/32	33/22
Seasons (O/D)	95/101	149/76	163/102	165/122	163/126	91/61
Mean WAR (O/D)	-0.355/+0.573	+0.421/+0.456	-0.140/+0.193	+0.139/+0.076	+0.246/+0.528	+0.572/+0.156
Difference (O - D)	-0.927	-0.035	-0.333	0.063	-0.282	0.416
U-statistic	3290	5389	7553	10275	9430	3151
p-value	0.0001	0.555	0.211	0.763	0.234	0.159
Significance	Highly Sig.	—	—	—	—	—

an apparent (but not statistically significant at typical confidence levels) edge, averaging +0.57 WAR per season compared to just +0.16 for defensive coaches. It is reasonable to assume that if the trend between offensive and defensive coaches holds for the remainder of the 2020's, that the difference in WAR would be statistically significant with the larger sample size.

The cumulative shift is clear: defensive coaches have declined by 0.41 WAR per season from the 1970s to 2020s, while offensive coaches have improved by 0.92 WAR per season over the same period, a complete reversal in relative performance. In today's NFL, offensive expertise appears to provide a meaningful competitive advantage that didn't exist in previous generations.

3.6 WAR Persistence and Scheme Obsolescence

Coaching performance is unstable from year to year. Analyzing 1,254 consecutive season pairs from 1970–2023, Year N coaching WAR explains only 7% of the variance in Year N+1 performance (Figure 7). Elite coaches averaging +2.64 wins above replacement the following season regress dramatically to just +0.65, losing 75% of their advantage. In contrast, the worst-performing coaches at -2.33 WAR improve substantially to -0.93 WAR when implementing a 10th percentile penalty for being terminated (Figure 8).

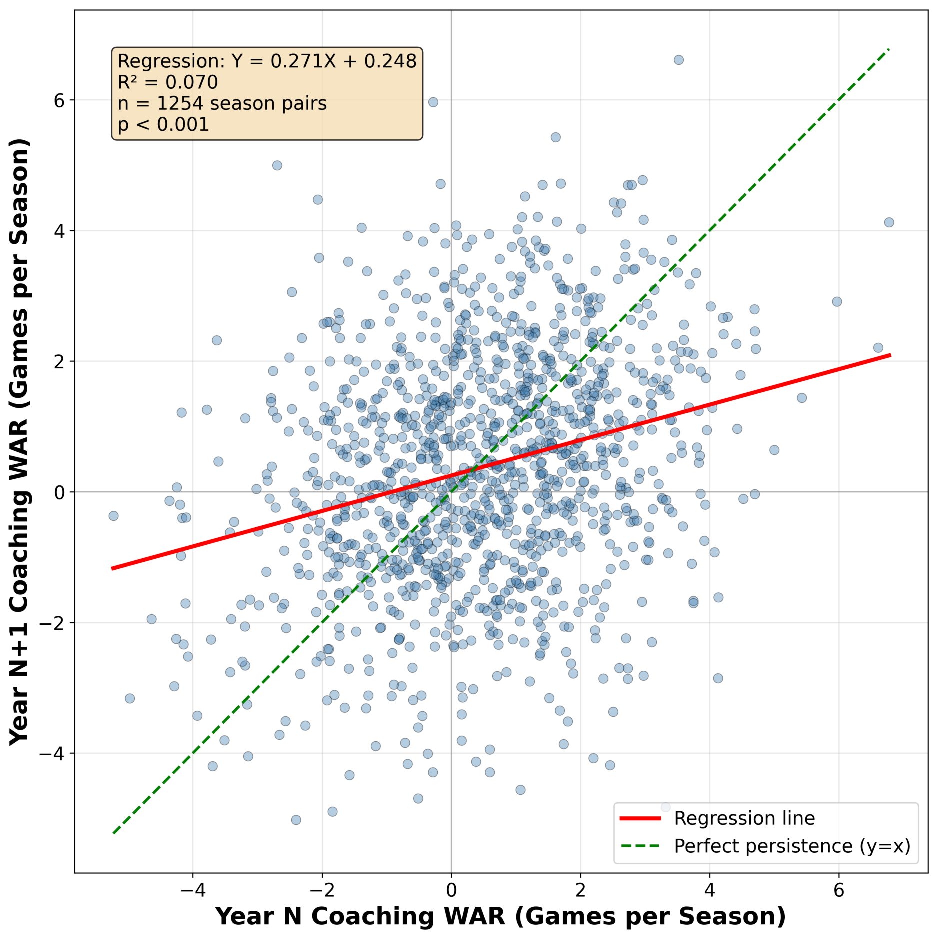

Fig. 7: Coach WAR in Year N+1 vs. Year N (persistence)

This volatility contrasts sharply with raw winning percentage, which shows nearly double the year-to-year stability ($R^2 = 0.137$, Appendix E, Figure 9). Winning percentage likely captures persistent organizational advantages, such as market size, ownership stability, accumulated roster talent, that carry forward regardless of coaching. The coach-specific contribution, by contrast, fluctuates, as what worked last season often doesn't work this season. This represents empirical evidence for scheme obsolescence.

But the instability isn't uniform across all coaches. While both offensive and defensive coaches show similar single-year persistence ($R^2 = 0.085$ and 0.064 respectively), multi-year averaging exposes a stark divergence. When evaluating coaches based on 2-year average performance, defensive coaches' predictive

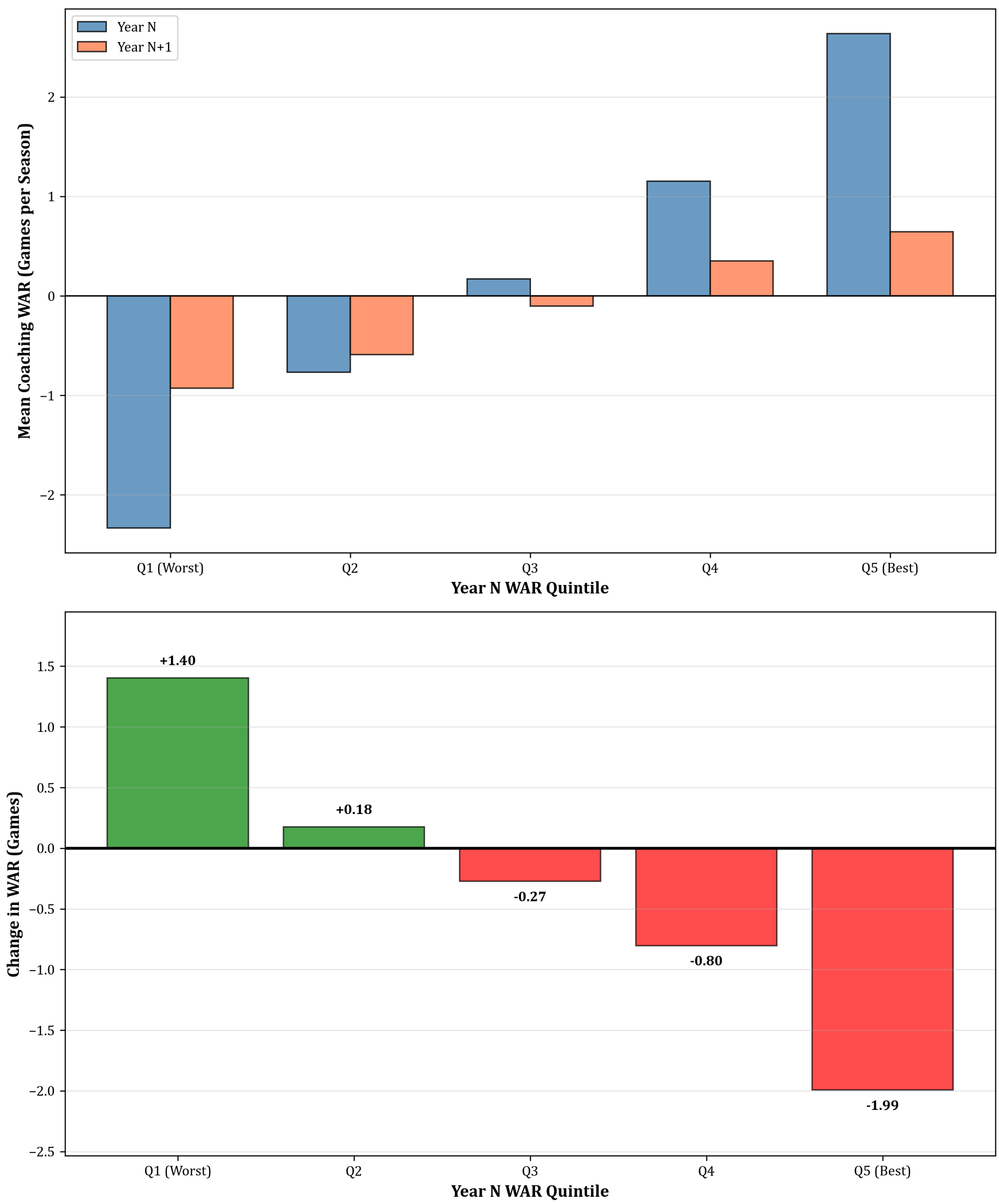

Fig. 8: Mean Coach WAR in Year N and Year N+1 by Quintile

power more than doubles to $R^2 = 0.131$. Offensive coaches' predictive power declines to $R^2 = 0.068$. This is a statistically significant difference at 90% confidence (Appendix F, Figure 10 and Table 12). At three-year horizons, the gap persists. The difference in persistence between Offensive and Defensive coaches highlights that Offensive schemes may have a shorter shelf life in the modern NFL. These patterns suggest fundamentally different retention strategies by coordinator background. Offensive coaches coming off elite seasons (+2.5 WAR) should trigger caution, not contract extensions, as their advantage will likely erode as schemes become known.

3.7 Limitations and the "Franchise Quarterback Effect"

Tables 7 and 8 show the top and bottom 10 single-season WAR instances in the modern era.

Tab. 7: Top 10 Single-Season WAR Instances

Rank	Team	Year	Coach	Actual Win %	Replacement Win %	WAR
1	CLT	2009	Jim Caldwell	0.875	0.451	+6.8
2	CHI	1987	Mike Ditka	0.733	0.320	+6.6
3	WAS	1982	Joe Gibbs	0.889	0.516	+6.0
4	NWE	2007	Bill Belichick	1.000	0.661	+5.4
5	CLT	1999	Jim Mora Sr	0.812	0.500	+5.0
6	NOR	2011	Sean Payton	0.812	0.514	+4.8
7	DET	2024	Dan Campbell	0.882	0.587	+4.7
8	RAI	1982	Tom Flores	0.889	0.594	+4.7
9	MIA	1974	Don Shula	0.786	0.492	+4.7
10	SFO	1987	Bill Walsh	0.867	0.573	+4.7

Tab. 8: Bottom 10 Single-Season WAR Instances

Rank	Team	Year	Coach	Actual Win %	Replacement Win %	WAR
1	SEA	1976	Jack Patera	0.143	0.470	-5.2
2	NOR	1996	Jim Mora Sr	0.188	0.501	-5.0
3	SEA	1992	Tom Flores	0.125	0.436	-5.0
4	CHI	2024	Matt Eberflus	0.294	0.600	-4.9
5	CLT	1991	Ron Meyer	0.062	0.364	-4.8
6	CAR	2001	George Seifert	0.062	0.356	-4.7
7	DET	2001	Marty Mornhinweg	0.125	0.415	-4.6
8	ATL	2020	Dan Quinn	0.250	0.535	-4.6
9	RAI	1997	Joe Bugel	0.250	0.527	-4.4
10	WAS	1994	Norv Turner	0.188	0.460	-4.4

The highest single-season coaching WAR values reveal an important limitation of roster-adjusted evaluation. In these extreme performances, three of the top six single-season WAR values belong to coaches during years with elite quarterback play (Peyton Manning in 1999, 2009, and Tom Brady in 2007). This "franchise quarterback effect" represents an inherent challenge in isolating coaching contributions: elite quarterbacks and elite coaching are rarely independent. While our framework controls for roster quality through metrics like player age, experience, and positional spending, it cannot fully disentangle whether a coach maximizes quarterback talent or whether an elite quarterback makes ordinary coaching appear exceptional. However, our framework's ability to identify coaching excellence in the absence of elite QB play (Dan Campbell 2024, Mike McDaniel with Tua, defense-focused teams in the 1970s and 1980s) suggests it captures more than just QB quality. Moreover, career-average analysis naturally controls for this.

At the opposite end of the spectrum, the poorest single-season performances ranged from Jack Patera's 1976 expansion Seahawks (-5.2 WAR) to Matt Eberflus's 2024 Bears (-4.9 WAR). Eberflus's Bears finished 5–12, yet the model predicted a replacement-level coach would achieve 9.6 wins with Chicago's roster and schedule.

The $\sim$12 win spread between the best and worst single-season performances highlights coaching's potential impact, even as the franchise quarterback effect reminds us that extreme values should be interpreted with appropriate caution.

Additional methodological limitations warrant consideration. First, while the model controls for roster quality through age, experience, turnover, and salary allocation, these proxy measures cannot fully capture player talent. Teams with similar demographic profiles may contain vastly different talent levels. Second, organizational factors beyond our data, such as ownership stability, drafting competence, and facility quality, likely influence outcomes but remain unobserved.

4 Conclusion

Head coaching represents one of the most consequential investments in professional sports. This study quantifies what NFL decision-makers have long struggled to measure: coaching quality, which fundamentally alters franchise trajectories independent of roster talent.

The practical implications are immediate and substantial. Front offices can now evaluate coaching candidates and incumbent coaches with circumstance-adjusted metrics that separate coaching impact from inherited advantages. Kevin O'Connell's +2.4 WAR per season with Minnesota demonstrates impressive coaching performance despite a roster that hasn't been considered exceptional. Conversely, Matt Eberflus's year 2 cumulative WAR of -3.9 could have prompted the organization to move on and avoid a lame-duck situation for Caleb Williams' rookie year.

The evolution of coaching value by coordinator background tells an equally compelling story about the modern NFL. Defensive-minded coaches dominated from the 1970s through the 1990s, but offensive coaches now generate 0.41 more WAR per season in the 2020s. Organizations hiring head coaches today should weigh offensive expertise more heavily than historical precedent would suggest, as the game itself has fundamentally shifted. Yet year-to-year performance analysis reveals this offensive advantage erodes rapidly: while defensive coaches' predictive power more than doubles when evaluated over multiple seasons, offensive coaches' predictive power declines by 20% over the same timeframe. This asymmetry suggests offensive schemes depreciate while defensive skills compound, a market dynamic with direct implications for retention decisions.

As NFL teams increasingly embrace analytics for player evaluation, from Expected Points Added to catch probability models, coaching assessment has remained surprisingly subjective, relying on win-loss records that confound coaching quality with roster talent, or playoff success that introduces large small-sample noise. But the stakes are too high for subjective evaluation: a single coaching decision can swing a franchise's trajectory by 5+ wins per season, on average, and by $\sim$12 wins in extreme circumstances.

The key question facing NFL organizations is whether they will continue to make eight-figure coaching decisions based on intuition and non-context-adjusted performance, or whether they will embrace objective, context-adjusted evaluation that separates true coaching impact from the circumstances coaches inherit. This Coach WAR framework provides the objective baseline the industry needs, quantifying the wins directly attributable to coaching excellence or deficiency, isolated from roster advantages, schedule strength, and organizational resources. Front offices armed with circumstance-adjusted coaching metrics can make these critical decisions with the same analytical rigor they now apply to draft picks and free agent signings, and the franchises that adopt it first will gain a sustainable competitive advantage in the league's most impactful personnel decision.

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A Feature Count Across Categories

Tab. 9: Feature Count Across Categories

Feature Category	Coach Feature?	Number of Features
Metadata	No	2
Salary Cap	No	19
Roster Turnover	No	63
Starter Turnover	No	27
Injury/Availability	No	37
Player Demographics	No	44
Penalty/Opponent	No	7
Draft Capital	No	29
Coaching Experience	Yes	7
Prior Team Performance (OC)	Yes	33
Prior Team Performance (DC)	Yes	33
Prior Team Performance (HC)	Yes	66
Target Variable	No	1
Total		**368**

B Selected Hyperparameters for the Final Model

Tab. 10: Selected Hyperparameters for the Final Model

Hyperparameter	Value
Objective	reg:squarederror
Random State	42
Number of Estimators	250
Learning Rate	0.05
Max Estimator Depth	3
Gamma	0
Lambda	1.5
Alpha	0.1
Subsample	1.0
Colsample by Tree	1.0
Minimum Child Weight	8

C Feature Importance Ranking of Top 20 Features

Tab. 11: Feature Importance Ranking of Top 20 Features

Rank	Feature Description	Importance	Coach Metric
1	Normalized first downs allowed under HC	0.050	Yes
2	Normalized opponent interceptions thrown	0.032	No
3	Normalized yards allowed under HC	0.028	Yes
4	Normalized team interceptions thrown	0.027	No
5	Reserve/injured/suspended cap as percentage	0.025	No
6	Normalized points allowed under HC	0.025	Yes
7	Average percentage of games missed by QBs	0.019	No
8	Normalized points scored under HC	0.019	Yes
9	Average experience of entire roster	0.013	No
10	Average experience of all offensive linemen	0.012	No
11	Number of new QBs added to roster	0.012	No
12	Normalized scoring percentage under HC	0.012	Yes
13	Normalized opponent avg yards per drive under HC	0.011	Yes
14	Active 53-man roster cap as percentage	0.011	No
15	Average experience of all starters	0.011	No
16	Minimum percentage of games missed by D-line	0.010	No
17	Average age of all starters	0.010	No
18	Normalized red zone attempts under HC	0.010	Yes
19	Normalized opponent avg points per drive under HC	0.010	Yes
20	Normalized yards per play allowed under HC	0.008	Yes

D Win Percentage Persistence

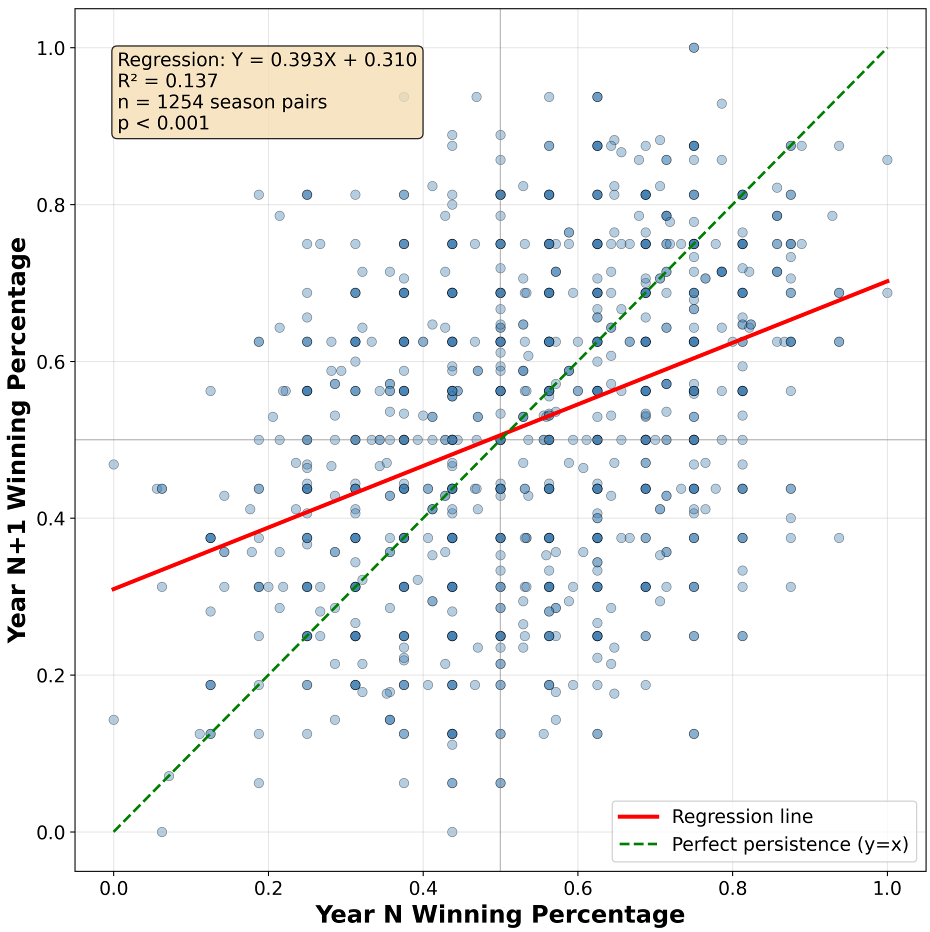

Fig. 9: Coach Win Percentage in Year N+1 vs. Year N

E WAR Persistence by Coach Background

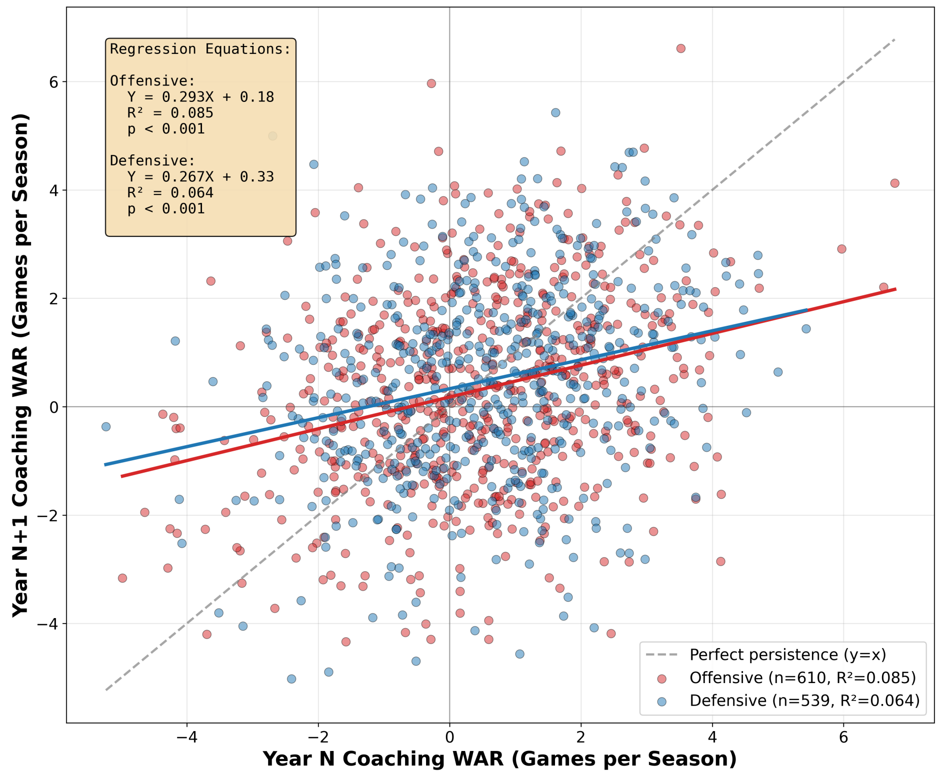

Fig. 10: Coach WAR in Year N+1 vs. Year N by Coach Background

Tab. 12: Coaching WAR Persistence by Background and Time Horizon

	Offensive		Defensive		Other	
Time Horizon	R^2	p-value	R^2	p-value	R^2	p-value
1-year	0.085	$<$0.001	0.064	$<$0.001	0.026	0.102
2-year avg	0.068	$<$0.001	0.131	$<$0.001	0.026	0.134
3-year avg	0.065	$<$0.001	0.119	$<$0.001	0.019	0.256
4-year avg	0.062	$<$0.001	0.095	$<$0.001	0.033	0.188

Note: Sample sizes for Offensive/Defensive/Other backgrounds: 1-year (610/539/105), 2-year avg (468/434/89), 3-year avg (346/332/69), 4-year avg (268/277/54).